

A First Course in Causal Inference Ch 7

Kodai Minesaki

Faculty of Engineering, The University of Tokyo

September 30, 2026

Ch. 7 Matched-Pairs Experiment

Ch. 7 Matched-Pairs Experiment

- ▶ The matched-pairs experiment (MPE) is the most extreme version of the SRE with only one treated unit and one control unit within each stratum.
- ▶ It has its own estimation and inference strategy

7.1 Design of the experiment and potential outcomes

- ▶ Consider an experiment with $2n$ units.
- ▶ If we have predictive covariates to the outcomes, we can pair units based on the similarity of covariates.
- ▶ With many covariates, we can define pairwise distances between units and the form pairs based on these distances.
- ▶ We assume that the pairs are formed based on the covariates and the discuss the subsequent design and analysis issues.

7.1 Design of the experiment and potential outcomes

Setup

- ▶ Let (i, j) index the unit j in pair i , where $i = 1, \dots, n$ and $j = 1, 2$.
- ▶ Unit (i, j) has potential outcomes $Y_{ij}(1)$ and $Y_{ij}(0)$ under the treatment and control, respectively.
- ▶ Within each pair, we randomly assign one into the treatment group ($Z_i = 1$) and the control group ($Z_i = 0$).

7.1 Design of the experiment and potential outcomes

Definition 7.1 (MPE)

The treatment assignments satisfy

$$(Z_i)_{i=1}^n \stackrel{\text{i.i.d.}}{\sim} \text{Bernoulli}\left(\frac{1}{2}\right).$$

The observed outcomes within pair i are

$$Y_{i1} = Z_i Y_{i1}(1) + (1 - Z_i) Y_{i1}(0) = \begin{cases} Y_{i1}(1), & \text{if } Z_i = 1, \\ Y_{i1}(0), & \text{if } Z_i = 0, \end{cases}$$

and

$$Y_{i2} = Z_i Y_{i2}(0) + (1 - Z_i) Y_{i2}(1) = \begin{cases} Y_{i2}(0), & \text{if } Z_i = 1, \\ Y_{i2}(1), & \text{if } Z_i = 0. \end{cases}$$

So the observed data are $(Z_i, Y_{i1}, Y_{i2})_{i=1}^n$

References I